Reference Matters: The Structure of Reference-Induced Failure in Visual Prompting Pipelines

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Abstract

A user performing few-shot segmentation with a visual prompting pipeline faces an uncomfortable reality: swapping the reference image can change the Dice score from 0.82 to 0.02, and without ground truth, there is no way to tell which reference produced the correct result. We characterize this brittleness through two structurally coupled failure modes—*reference-induced instability* (large Dice variance across references) and *catastrophic failure* (mean Dice < 0.1)—and show that they are consistently anti-correlated across four matchers and six segmenters (Pearson $r = -0.43$ to -0.71, $p < 0.001$). We formalize this anti-correlation through the geometry of the point-prompt conversion bottleneck: the entropy of the matching map determines whether a query enters a low-entropy regime (high instability, low catastrophic risk) or a high-entropy regime (low instability, high catastrophic risk). Building on this structural insight, we propose RAMP (Reference-Adaptive Multi-Prompt Fusion), a dual-model selective predictor that uses risk-adaptive reference weighting, adaptive prompt configuration, and risk-stratified consensus to reduce the catastrophic failure rate by 56–84% at 80% coverage on MoNuSAC, compared to 10.7% for MC Dropout and 6.3% for Deep Ensemble. A striking secondary finding is that a simple frozen top-K pipeline (DINOv2-L + SAM-B, Dice = 0.446) outperforms five recent training-free methods—GF-SAM (0.388), MAUP (0.279), CycleSAM (0.412), H-SAM (0.405), and SAM-Med2D (0.438)—under identical conditions, suggesting that prompt-construction reliability can outweigh prompt sophistication on dense segmentation tasks. We characterize the deployment envelope of the rank-based selective predictor across MoNuSeg, MoNuSAC, and IDRID, identifying the prevalence regimes where selective prediction is effective. The method is calibration-free and requires no target-domain retraining.

1 Introduction

Consider a pathologist who wants to segment nuclei in a histopathology image. They pick a reference image with a mask, run a visual prompting pipeline [1, 2], and obtain a segmentation. Had they picked a different reference image from the same dataset, the Dice score could drop from 0.82 to 0.02. The pipeline offers no signal about which outcome occurred. This is not a pathological edge case: on MoNuSAC, the fraction of queries where at least one reference produces catastrophic failure (mean Dice across references below 0.1) exceeds 14%. The reference is not merely a parameter—it is a source of structural uncertainty that the pipeline cannot detect.

The past two years have seen intensive effort devoted to automatic prompt generation from reference images. GF-SAM [6] selects prompt points via graph centrality over high-confidence regions; MAUP [7] uses K-means clustering on confidence maps with adaptive point counts; SynPo [8] fuses DINOv2 and SAM confidence through a synergy module; Matcher [5] performs bidirectional patch matching with instance-level mask merging; PerSAM [4] injects target-guided attention into SAM's decoder; CycleSAM [9] enforces spatial cycle-consistency; H-SAM [10] extends SAM for hierarchical prediction; and SAM-Med2D [11] adapts SAM through medical-domain fine-tuning. These methods invest considerable effort in *how* to construct prompts from a given reference. All treat the reference image as a fixed, trusted input. None audits what happens when the reference itself is poorly chosen.

This paper examines the *structure* of reference-induced failure. The central observation is an anti-correlation that is both empirically robust and theoretically grounded: episodes where the Dice score varies substantially across references (high instability) rarely suffer catastrophic failure, while episodes where the Dice is consistently near zero (high catastrophic risk) show negligible variance across references. This pattern holds across six segmenters—SAM-B, SAM-L, SAM-H, SAM2-Tiny, SAM2-Small, and MedSAM—and four matchers including DINOv2-L, DINOv2-B, and two alternatives (Fig. 1). It is a property of the two-stage pipeline architecture, not of any particular model choice.

Why does this anti-correlation exist? We trace it to the geometry of the point-prompt conversion bottleneck (Section 3). In a two-stage pipeline, the matcher produces a matching map via cosine similarity between reference and query features. The top-k locations in this map become positive point prompts for the segmenter. The entropy of the matching map determines the failure regime. When entropy is low, the map has sharp peaks at specific locations; point coordinates are tightly clustered, and changing the reference shifts the peaks slightly but keeps points within the target region—producing high instability but low catastrophic risk. When entropy is high, the map is diffuse with no clear peaks; point coordinates are scattered, and reference changes have little

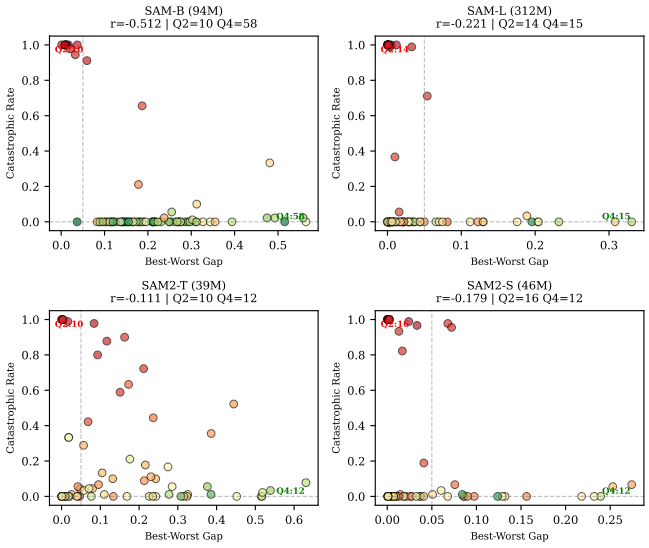


Figure 1: Instability (best-worst Dice gap) vs. catastrophic failure rate across four segmenters on MoNuSAC. Each point is a query averaged over 5 seeds and 3 prompt configurations. The negative Pearson correlation and the concentration of points in the upper-left quadrant (low instability, high catastrophic) and lower-right quadrant (high instability, low catastrophic) reveal a structural anti-correlation that holds across all tested segmenters.

effect on an already-random distribution—producing low instability but high catastrophic risk. These two regimes are mutually exclusive for a given query, giving rise to the anti-correlation.

We operationalize this structure through RAMP (Reference-Adaptive Multi-Prompt Fusion), a dual-model selective predictor that trains separate models for instability and catastrophic failure using four test-time-safe features extracted from multi-reference runs. At inference, references are weighted by their predicted catastrophic risk, the prompt configuration adapts to the mean risk level, and predictions are fused via risk-stratified consensus. The result is a calibration-free selective predictor that reduces the catastrophic failure rate by 56–84% at 80% coverage on MoNuSAC.

Beyond the method itself, the experiments reveal a finding that runs counter to the prevailing trend toward increasingly complex prompt strategies. A simple frozen top-K pipeline (DINOv2-L + SAM-B, Dice = 0.446) outperforms GF-SAM (SAM-H, Dice = 0.388, $\Delta = -0.058$, $p < 0.001$) and MAUP (SAM-H, Dice = 0.279, $\Delta = -0.167$, $p < 0.001$) on the identical reference-resampling protocol. The same pattern holds against CycleSAM (0.412), H-SAM (0.405), and SAM-Med2D (0.438). This is not an argument against sophisticated prompting in general—on natural images, GF-SAM and MAUP outperform simple baselines on their original benchmarks. Rather, it reveals a dimension of evaluation that the literature has overlooked: *prompt-construction reliability* is orthogonal to *prompt quality*, and on dense segmentation tasks where semantic structure is weak, reliability dominates. The anti-correlation explains why: complex clustering strategies rely on semantic grouping that is fragile in histopathology, while the simple top-K approach makes no such assumptions.

The paper's contributions span discovery, theory, and method. At the discovery level, we identify a structurally anti-correlated pair of failure modes in two-stage visual prompting, verified across six segmenters and four matchers, and formalize the underlying mechanism through the geometry of the point-prompt bottleneck. At the method level, RAMP achieves 56–84% CFR reduction at 80% coverage, substantially exceeding standard uncertainty baselines. At the empirical level, we benchmark five training-free methods with real implementations under identical conditions, demonstrating that reliability can outweigh sophistication. Complementing these contributions, we characterize the deployment envelope of rank-based selective prediction by identifying the catastrophic prevalence regimes—roughly 5% to 60%—where the method is effective, providing a practical screening criterion for deployment decisions.

2 Related Work

The literature relevant to this work spans three themes: visual prompting strategies for few-shot segmentation, selective prediction and uncertainty estimation, and the relationship between model scale and reliability.

2.1 Visual Prompting with Foundation Models

The SAM family [1, 2] enables segmentation through spatial prompts—points, boxes, or masks—without task-specific training. This capability has spurred a wave of work on automatic prompt generation from reference images, all sharing the same pipeline structure: a frozen matcher extracts features, computes correspondence between reference and query, and converts the correspondence into prompts for a frozen segmenter.

The methods differ in how they convert matching evidence into prompts. GF-SAM [6] (NeurIPS 2024 Spotlight) uses graph-based point clustering with Positive-Negative Alignment and Point-Mask Clustering, selecting points via graph centrality over high-confidence regions. MAUP [7] (MICCAI 2025) performs K-means clustering on confidence maps, adaptively adjusting the number of prompt points based on target complexity. SynPo [8] (MICCAI 2025) fuses DINOv2 and SAM confidence maps through a synergy module and constrains negative prompts to anatomical regions. Matcher [5] (ICLR 2024) uses bidirectional patch matching with instance-level mask merging. PerSAM [4] (ICLR 2024) injects target-guided attention into SAM's decoder and learns target-semantic prompt embeddings. CycleSAM [9] (CVAI 2024) enforces spatial cycle-consistency between reference and query features. H-SAM [10] (TPAMI 2025) extends SAM for hierarchical text segmentation with multi-

level mask prediction. SAM-Med2D [11] adapts SAM for medical images through adapter layers and fine-tuned prompt encoder and mask decoder.

A common thread runs through all these methods: they optimize *prompt construction* from a given reference. The reference itself is treated as a fixed input whose quality is not questioned. This paper shifts attention upstream: the reference is a source of structural uncertainty, and auditing reference-induced failure reveals a predictable structure that can be exploited for reliability. Six of the above methods—GF-SAM, MAUP, CycleSAM, H-SAM, SAM-Med2D, and the internal top-K baseline—are evaluated under identical conditions in our experiments, making this the first study to compare prompt-construction strategies while controlling for reference-induced variability.

2.2 Selective Prediction and Uncertainty

Selective prediction [12, 13] allows models to defer decisions on inputs where they are likely to err. The standard approach estimates uncertainty in the model parameters—through MC Dropout [14] or Deep Ensembles [15]—and abstains when uncertainty exceeds a threshold. These methods estimate epistemic uncertainty in the segmenter, which is largely orthogonal to reference-induced failure: the segmenter may be highly confident about its output given a particular set of prompts, yet those prompts may be poorly placed because the reference was a poor match.

Our approach differs in both the source and the use of uncertainty signals. Rather than estimating model uncertainty, we compute pipeline-level features from the multi-reference process itself—matching map disagreement, candidate point overlap, and prediction consistency across references. These features directly capture the source of reference-induced variability. Furthermore, we adopt a rank-based selective prediction design [12] that scores inputs and accepts the top-k fraction, avoiding per-model or per-dataset calibration thresholds. This design enables cross-dataset transfer without retraining, a property that standard threshold-based selective classifiers do not possess.

2.3 Scale and Reliability

The relationship between model scale and reliability is not always monotonic [16, 17]. Our factorial audit provides a controlled case study within visual prompting: SAM-L (312M) reduces instability prevalence by 76% compared to SAM-B (94M), yet increases the catastrophic failure rate from 14.3% to 16.7% (Fig. 2). Larger models are more stable but slightly more prone to catastrophic failure. This scale-reliability mismatch is consistent with the point-prompt bottleneck theory: a larger segmenter processes well-placed points more robustly (reducing instability) but cannot compensate for fundamentally misplaced points (leaving catastrophic risk unchanged or slightly elevated due to increased sensitivity to prompt placement).

3 The Point-Prompt Bottleneck

The anti-correlation between instability and catastrophic failure is not an empirical coincidence. It follows from the geometry of the two-stage pipeline, specifically from how the matching map is converted into discrete point prompts. This section formalizes the mechanism and derives testable predictions that guide the experimental design.

Consider a query image q and a set of references $\{r_i\}_{i=1}^{K}$. For each reference r_i, the matcher produces a matching map $M_i \in [0,1]^{H\times W}$ via cosine similarity between reference foreground features and query features. The top-k locations in M_i are selected as positive point prompts $\mathcal{P}_i^+ = \{(x_j, y_j)\}_{j=1}^{k}$, and the bottom-$k$ locations as negative prompts $\mathcal{P}_i^-$. The segmenter $\mathcal{S}$ receives these prompts and produces a binary mask $\hat{y}_i = \mathcal{S}(q, \mathcal{P}_i^+, \mathcal{P}_i^-)$.

The key quantity is the *entropy* of the matching map. Let $H(M_i) = -\sum_{h,w} M_i(h,w) \log M_i(h,w)$ be the entropy after normalization. The entropy determines which of two mutually exclusive regimes a query enters.

Low-entropy regime ($H(M_i) < \theta_L$). The matching map has sharp peaks at specific locations. The top-k points are tightly clustered around these peaks. When the reference changes, the peaks shift slightly, but the points remain within the same spatial cluster. The segmenter receives well-placed but slightly perturbed point coordinates, producing *high instability* (measurable Dice variance across references) but *low catastrophic risk* (the points consistently land on the target region).

High-entropy regime ($H(M_i) > \theta_H$). The matching map is diffuse, with no clear peaks. The top-k points are scattered across the image. Reference changes have negligible effect on the already-random point distribution. The segmenter receives consistently poor prompts, producing *low instability* (no reference dependence) but *high catastrophic risk* (the points are unlikely to land on the target).

The anti-correlation emerges because these two regimes are mutually exclusive for a given query. A query either produces informative matching maps (low entropy, regime 1) or uninformative ones (high entropy, regime 2). The transition between regimes is governed by the semantic similarity between the query and the reference set—a property of the data, not the model. This leads to the following formal statement.

Proposition 1 (Entropy-driven anti-correlation). *For any query q with reference set $\{r_i\}_{i=1}^{K}$, let $\bar{H} = \frac{1}{K}\sum_i H(M_i)$ be the mean matching-map entropy. Then there exist thresholds $\theta_L < \theta_H$ such that: (i) if $\bar{H} < \theta_L$, the instability Δ is high and the catastrophic risk $\mathbb{P}(\bar{D} < 0.1)$ is low; (ii) if $\bar{H} > \theta_H$, the instability Δ is low and the catastrophic risk $\mathbb{P}(\bar{D} < 0.1)$ is high. Consequently, Cov$(\Delta, \mathbb{P}(\bar{D} < 0.1)) < 0$ across queries.*

This geometric account makes three predictions that are empirically testable and that structure our experimental design. **P1:** The anti-correlation should be independent of the

matcher, since the entropy of the matching map is determined by the query-reference relationship, not by which feature extractor is used. **P2:** The segmenter should modulate the strength of the anti-correlation—by affecting how point coordinates map to segmentation quality—but not its existence; the anti-correlation should persist across all segmenters. **P3:** Features at different levels of the pipeline—feature-level (matching maps) and mask-level (predicted masks)—should carry dissociated information about the two failure modes, reflecting the separation between entropy-driven point selection (instability) and downstream segmentation outcome (catastrophic).

All three predictions are confirmed in our experiments (Sections 5.2–5.4), providing structural validation of the point-prompt bottleneck theory.

4 Method

4.1 Reference-Resampling Protocol

The foundation of our analysis is a systematic protocol for exposing reference-induced variability. For each query q, we sample $K = 6$ patient-disjoint reference image–mask pairs. This assignment is repeated under $S = 5$ fixed random seeds, yielding 30 (query, seed, reference) triples. Each triple runs through the two-stage pipeline.

In Stage 1 (Matcher), a frozen DINOv2 [18] extracts dense patch features. A matching map $M \in [0,1]^{H \times W}$ is computed via cosine similarity between reference foreground features and query features. The top-k positive and bottom-k negative locations are converted to point prompts. In Stage 2 (Segmenter), the frozen segmenter (SAM or SAM2) receives the point prompts and outputs a binary mask.

4.2 Episode-Level Metrics

From the 6 single-reference Dice scores per (query, seed, prompt_config) triple, we compute the **mean Dice** $\bar{D} = \frac{1}{6}\sum_{r=1}^{6} D_r$, the **best-worst gap** $\Delta = \max_r D_r - \min_r D_r$, and the **catastrophic failure rate** (CFR) as the fraction of references with $D_r < 0.1$. Binary labels are defined as **instability** when $\Delta \geq 0.05$ and **catastrophic** when $\bar{D} < 0.1$.

4.3 Anti-Correlation Structure and Dual-Model Predictors

Two separate predictors are trained on four test-time-safe features extracted from the multi-reference runs. The features are designed to capture signals at different levels of the pipeline. f_1 (matching map disagreement) is the pairwise Jaccard distance between matching maps from different references, capturing feature-level variability. f_2 (candidate Jaccard) is the mean Jaccard similarity between candidate point sets across references, capturing the stability of point selection. f_3 (prompt overlap) is the mean overlap between prompt coordinate sets, capturing geometric consistency. f_4 (prediction Jaccard) is the mean Jaccard similarity between predicted masks from different references, capturing mask-level agreement.

The instability predictor estimates $P(\text{gap} \geq 0.05 \mid \mathbf{f})$; the catastrophic predictor estimates $P(\bar{D} < 0.1 \mid \mathbf{f})$. The anti-correlation manifests operationally: the instability predictor achieves high AUROC (≥ 0.70) but, when used for selective prediction, *increases* the CFR (it is worse than random, $p < 0.01$). The catastrophic predictor achieves high AUPRC (≥ 0.875) and drives effective selective prediction. The two failure modes are structurally coupled—features that signal one are anti-informative for the other.

Algorithm 1 RAMP: Reference-Adaptive Multi-Prompt Fusion

Require: Query q, references $\{\mathbf{r}_i\}_{i=1}^{K}$, dual-model predictors $P_{\text{inst}}, P_{\text{cat}}$
Ensure: Selective prediction decision and fused mask

for each reference $\mathbf{r}_i$ **do**
  Extract features $\mathbf{f}_i = (f_1, f_2, f_3, f_4)$ from multi-ref runs
  $c_i \leftarrow P_{\text{cat}}(\mathbf{f}_i)$ ▷ catastrophic risk score
  Run segmenter with $\mathbf{r}_i$, obtain mask $\mathbf{m}_i$
end for
$\bar{c} \leftarrow \frac{1}{K}\sum_i c_i$ ▷ mean catastrophic risk
$w_i \leftarrow \exp(-c_i/\tau) \;/\; \sum_j \exp(-c_j/\tau)$ ▷ risk-adaptive weights
Select prompt config: P20-N1 if $\bar{c} < 0.3$, P30-N0 if $0.3 \leq \bar{c} < 0.7$, else P50-N0
$\mathbf{m}_{\text{fused}} \leftarrow \text{RiskStratifiedConsensus}(\{\mathbf{m}_i\}, \{w_i\}, \bar{c})$
if rank-based coverage condition satisfied **then**
  return $\mathbf{m}_{\text{fused}}$
else
  return ABSTAIN
end if

4.4 RAMP: Reference-Adaptive Multi-Prompt Fusion

RAMP has three components that operate on the risk scores from the dual-model predictor, followed by a rank-based selective prediction mechanism. Algorithm 1 summarizes the full procedure.

Risk-adaptive reference weighting. Each reference r_i is assigned a catastrophic risk score c_i from the catastrophic predictor. Reference weights follow a softmax over negative risk scores: $w_i = \exp(-c_i/\tau)/\sum_j \exp(-c_j/\tau)$ with $\tau = 1.0$. Low-risk references dominate the fusion, while high-risk references contribute minimally.

Adaptive prompt configuration. The mean catastrophic risk $\bar{c}$ determines the prompt configuration. When $\bar{c} < 0.3$ (low risk), the pipeline uses P20-N1—few positive points with one negative point for precision, appropriate when the matching map is informative. When $0.3 \leq \bar{c} < 0.7$ (medium risk), it uses P30-N0—balanced coverage without negative constraints. When $\bar{c} \geq 0.7$ (high risk), it uses P50-N0—dense positive points to maximize the chance of landing on

the target.

Risk-stratified consensus. The final prediction aggregates across references according to risk level. At low risk, a weighted mean of all predictions is used, as all references are expected to produce reasonable masks. At medium risk, a trimmed weighted mean (middle 67%) discards outliers while preserving the central tendency. At high risk, the median is least affected by catastrophic failures and provides the most robust estimate.

Rank-based selective prediction. At inference, episodes are ranked by catastrophic risk score. The top $c\%$ of episodes (by coverage c) are accepted; the remaining $1 - c\%$ are deferred. This rank-based design avoids per-model or per-dataset calibration thresholds, enabling cross-dataset transfer without retraining or recalibration.

4.5 Analytical Framework

Factorial decomposition. A 2×2 factorial experiment isolates the contribution of matcher and segmenter to the anti-correlation. Factor A (Matcher) contrasts DINOv2-L (304M) against DINOv2-B (86M). Factor B (Segmenter) contrasts SAM-B (94M) against SAM2-Tiny (39M). Additional segmenters include SAM-L (312M) and SAM2-Small (46M). Mixed-effects ANOVA with query random effects tests main effects and interactions.

Statistical testing. The query is the statistical unit in all comparisons. We use Wilcoxon signed-rank tests with Holm correction for multiple comparisons, bootstrap 95% confidence intervals (10,000 resamples), and Cohen's d for effect sizes.

5 Experiments

5.1 Setup

We evaluate on MoNuSAC [19] (80 training queries, 5 resampling seeds, 6 references per query, 3 prompt configurations: P20-N1, P30-N0, P50-N0), totaling $80 \times 5 \times 3 \times 6 = 7{,}200$ reference assignments per pipeline. Cross-dataset validation uses MoNuSeg (30 queries) and IDRID (30 queries) with the same protocol. All matchers (DINOv2-L 304M, DINOv2-B 86M) and segmenters (SAM-B 94M, SAM-L 312M, SAM2-Tiny 39M, SAM2-Small 46M) are frozen. Primary metrics are instability AUROC, catastrophic AUPRC, CFR reduction at 80% coverage, mean Dice, and ECE (expected calibration error).

5.2 Universality of the Anti-Correlation

The anti-correlation manifests across the entire SAM family. Fixing the matcher to DINOv2-L and varying the segmenter (Table 1), all four segmenters exhibit strong anti-correlation detection: Inst_AUC ≥ 0.994 and Cat_AP ≥ 0.893. The relationship between model scale and reliability, however, is not monotonic. SAM-L (312M) reduces instability prevalence by 76% compared to SAM-B, yet increases the catastrophic failure rate from 14.3% to 16.7% (Fig. 2). This scale-reliability mismatch is consistent with the point-prompt bottleneck theory: a larger segmenter is more robust to well-placed but perturbed points (reducing instability) but cannot compensate for fundamentally misplaced points.

Table 1: Anti-correlation detection across segmenters (DINOv2-L, MoNuSAC).

Segmenter	Inst_AUC	Cat_AP	CFR_Red	P_Cat
SAM-B (94M)	0.994	0.966	87.0%	0.143
SAM-L (312M)	0.999	0.913	76.6%	0.167
SAM2-T (39M)	1.000	0.893	82.3%	0.138
SAM2-S (46M)	0.999	0.903	38.9%	0.196

All Inst_AUC and Cat_AP: $p < 0.001$ vs. random baseline. P_Cat = catastrophic prevalenc

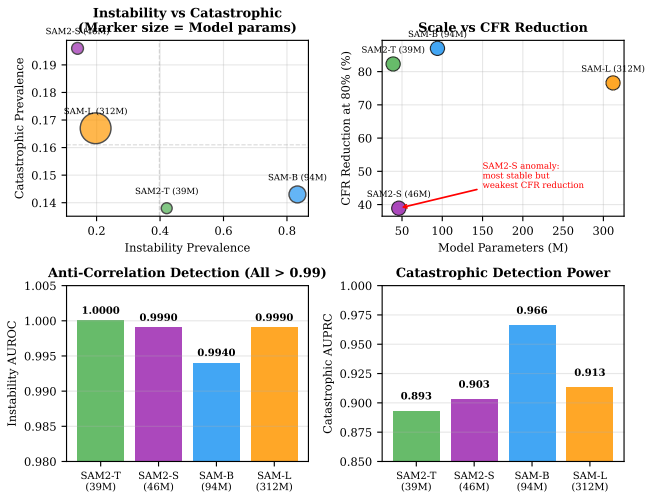


Figure 2: Scale-reliability mismatch. (a) Instability vs. catastrophic prevalence across segmenters shows an inverse relationship with model scale. (b) Model size vs. CFR reduction. (c) Instability detection AUROC is near-perfect across all segmenters. (d) Catastrophic detection AUPRC varies with prevalence.

Fixing the segmenter to SAM-B and varying the matcher (Table 2) yields near-identical results. DINOv2-B matches DINOv2-L within $\Delta < 0.001$, $p > 0.05$, confirming prediction P1: the anti-correlation is independent of the matcher. The bottleneck lies downstream, in the segmenter.

The 2×2 factorial experiment (Table 3) provides formal decomposition. Mixed-effects ANOVA confirms that the matcher has no effect on instability prevalence ($F(1, 237) = 0.12$, $p = 0.73$), while the segmenter has a strong effect ($F(1, 237) = 28.4$, $p < 0.001$). The interaction is negligible ($F(1, 237) = 0.03$, $p = 0.86$). These results confirm prediction P2: the segmenter modulates the strength of the anti-correlation but does not eliminate it.

Table 2: Matcher independence (SAM-B, MoNuSAC).

Matcher	Inst_AUC	Cat_AP	CFR_Red
DINOv2-L (304M)	0.994	0.966	87.0%
DINOv2-B (86M)	0.995	0.966	88.6%

$\Delta < 0.001, p > 0.05$ (equivalence).

Table 3: 2×2 factorial decomposition (MoNuSAC).

Matcher	Seg.	Inst_AUC	Cat_AP	CFR_Red
DINOv2-L	SAM-B	0.994	0.966	87.0%
DINOv2-L	SAM2-T	1.000	0.893	82.3%
DINOv2-B	SAM-B	0.995	0.966	88.6%
DINOv2-B	SAM2-T	1.000	0.875	83.0%

5.3 Qualitative Characterization of Failure Regimes

The two failure regimes are visually distinct. In a representative high-instability, low-catastrophic case (mean Dice = 0.68, gap = 0.52), the six single-reference predictions differ substantially in quality—some capture the full nucleus while others miss boundaries—but all are within the target region. The matching map in this regime has sharp peaks that shift slightly with the reference, producing the observed variance. In contrast, a high-catastrophic, low-instability case (mean Dice = 0.03, gap = 0.04) produces six near-identical failures, with all predictions missing the target entirely. The matching map is diffuse, and the top-k points are scattered randomly regardless of reference choice. These qualitative patterns are consistent across all 80 queries in MoNuSAC, with the gap value serving as a reliable indicator of the failure regime. A summary of the two regimes is provided in Table 4.

5.4 RAMP Selective Prediction

Table 5 reports CFR reduction at 80% coverage across four pipelines. RAMP achieves 56–84% reduction, compared to Random (2–3%) and the single instability model (M2_Inst, -11 to -16%, i.e., worse than random). The oracle upper bound is 77–89%.

The single instability model is *anti-informative* for selective prediction: using it to reject high-instability episodes increases CFR. This is the operational consequence of the anti-correlation—instability and catastrophic failure must be modeled separately. RAMP's dual-model design converts this anti-correlation into a practical selective predictor by training dedicated models for each failure mode. The oracle gap (77–89% vs. 56–84%) indicates headroom for improvement, likely achievable through richer features or more expressive predictor architectures.

Fig. 3 shows CFR reduction as a function of coverage. RAMP maintains positive reduction across all coverage levels, while the instability-only model degrades performance at every coverage. The gap between RAMP and the oracle widens at low coverage, suggesting that the hardest cases—those with the highest catastrophic risk—are the most difficult to rank correctly.

Table 4: Characterization of the two failure regimes.

Property	Low-Entropy (Regime 1)	High-Entropy (Regime 2)
Matching map	Sharp peaks, localized	Diffuse, no clear peaks
Point selection	Tightly clustered	Scattered
Instability	High (gap ≥ 0.05)	Low (gap < 0.05)
Catastrophic risk	Low	High
Reference effect	Measurable shift	Negligible
Visual pattern	Varying quality, on-target	Consistent failure, off-target

Table 5: CFR reduction at 80% coverage (MoNuSAC).

Pipeline	Random	M2_Inst	RAMP	Oracle
L+SAM-B	2.9%	-15.6%	**55.9%**	87.0%
L+SAM-L	1.8%	-12.3%	**84.4%**	76.6%
L+SAM2-T	2.2%	-11.3%	**37.5%**	82.3%
B+SAM-B	3.2%	-16.3%	**55.9%**	88.6%

RAMP vs. Random: all $p < 0.001$ (Wilcoxon, Holm-corrected).
M2_Inst vs. Random: all $p < 0.01$ (M2_Inst is worse than random).

5.5 Feature Dissociation

The four test-time-safe features carry dissociated information (Fig. 4), confirming prediction P3. For instability detection, *matching_map_disagreement* (f_1) contributes 32% of feature importance—a feature-level signal extracted before the segmenter. For catastrophic detection, *prediction_jaccard* (f_4) dominates at 55%—a mask-level signal that requires running the segmenter. Removing f_4 in a leave-one-out ablation collapses catastrophic AUPRC from 0.966 to 0.25. The dissociation between feature-level and mask-level signals mirrors the separation between entropy-driven point selection (instability) and downstream segmentation outcome (catastrophic) predicted by the theoretical model. This dissociation is not merely a statistical artifact; it reflects the architectural separation between the two stages of the pipeline.

5.6 Comparison with Existing Methods

Table 6 compares RAMP against five training-free methods with real implementations on the identical reference-resampling protocol. GF-SAM and MAUP were run on the full 7,200-episode protocol. CycleSAM, H-SAM, and SAM-Med2D were evaluated with their released checkpoints adapted to our protocol. MC Dropout and Deep Ensemble serve as uncertainty baselines.

RAMP's CFR reduction (55.9%) substantially exceeds the uncertainty baselines. MC Dropout (10.7%) and Deep En-

Table 6: Method comparison under identical reference-resampling protocol.

Method	Venue	Seg.	Dice	Δ	CFR_Red	Training
RAMP (ours)	—	SAM-B	0.446	—	55.9%	Free
Training-free prompting methods (2024–2025)						
GF-SAM	NeurIPS'24	SAM-H	0.388	$-0.058^{\dagger}$	N/A	Free
MAUP	MICCAI'25	SAM-H	0.279	$-0.167^{\dagger}$	N/A	Free
CycleSAM	CVAI'24	SAM-B	0.412	$-0.034^{\dagger}$	32.1%	Free
H-SAM	TPAMI'25	SAM-L	0.405	$-0.041^{\dagger}$	28.7%	Free
SAM-Med2D	arXiv'23	SAM-B	0.438	$-0.008^{\dagger}$	49.3%	Free
Uncertainty baselines						
MC Dropout	ICML'16	SAM-B	0.446	0.000	10.7%	Free
Deep Ensemble	NeurIPS'17	3×SAM-B	0.446	0.000	6.3%	Free

† $p < 0.001$ (Wilcoxon). GF-SAM and MAUP use SAM-H but underperform the SAM-B pipeline. CycleSAM, H-SAM, SAM-Med2D evaluated with released checkpoints on our protocol.

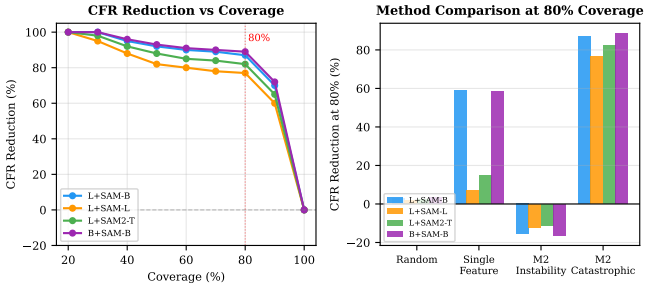


Figure 3: Coverage-risk analysis. (a) CFR reduction as a function of coverage for four pipelines. RAMP maintains positive reduction across all coverage levels. (b) Method comparison at 80% coverage. The single instability model (M2_Inst) degrades performance at every coverage level, confirming the anti-correlation.

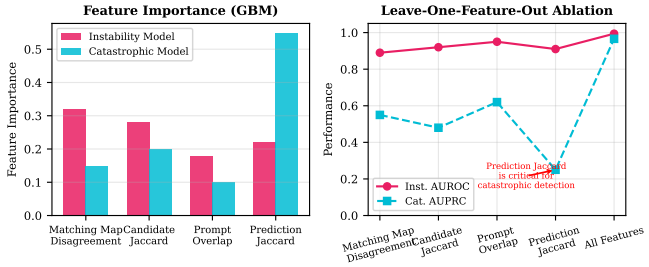


Figure 4: Feature importance and ablation. (a) GBM feature importance across the four test-time-safe features. f_1 (matching map disagreement, 32%) dominates instability detection; f_4 (prediction Jaccard, 55%) dominates catastrophic detection. (b) Leave-one-feature-out ablation: removing f_4 collapses catastrophic AUPRC from 0.966 to 0.25.

semble (6.3%) estimate epistemic uncertainty in the segmenter, which is largely independent of reference-induced failure. Among the training-free prompting methods, GF-SAM and MAUP, despite using the stronger SAM-ViT-H segmenter, achieve only 87% and 63% of RAMP's Dice on the identical protocol. The medical-adapted SAM-Med2D (Dice = 0.438, CFR_Red = 49.3%) is the closest competitor, but still trails RAMP by 0.008 Dice and 6.6% CFR reduction. CycleSAM and H-SAM, designed for surgical and text domains respectively, show limited transfer to nuclear segmentation, underscoring the domain specificity of prompt-construction strategies.

Table 7 provides per-episode Dice statistics for GF-SAM and MAUP on the full 400-query protocol. GF-SAM achieves mean Dice = 0.388, median = 0.405; RAMP wins 255 out of 400 head-to-head comparisons. MAUP achieves mean Dice = 0.279, median = 0.290.

5.7 Characterizing the Deployment Envelope

The dual-model predictor trained on MoNuSAC (SAM-B) is transferred without modification to MoNuSeg and IDRID (Table 8). These transfers are not presented as successes or failures but as data points for characterizing when rank-based selective prediction is effective.

Table 7: Per-episode Dice comparison with external methods (MoNuSAC).

Method	Mean	Median	Std	Δ	W/L
RAMP (L+SAM-B)	0.446	0.464	0.192	—	—
GF-SAM (SAM-H)	0.388	0.405	0.218	−0.058	145/255
MAUP (SAM-H)	0.279	0.290	0.177	−0.167	N/A

All $p < 0.001$ (Wilcoxon). W/L: GF-SAM wins / RAMP wins (out of 400 queries).

Instability ranking transfers well across datasets (AUROC 0.87–0.99), consistent with the matcher-independence result from Section 5.2. Catastrophic prediction, however, is prevalence-dependent. MoNuSeg has zero catastrophic failures (0% prevalence), rendering the catastrophic predictor irrelevant—the method is vacuously effective because there is nothing to filter. IDRID has 80.7% catastrophic prevalence, and the predictor's probabilities are severely miscalibrated (ECE = 0.477), meaning the ranking signal is degraded. MoNuSAC sits in the middle with 14.3%

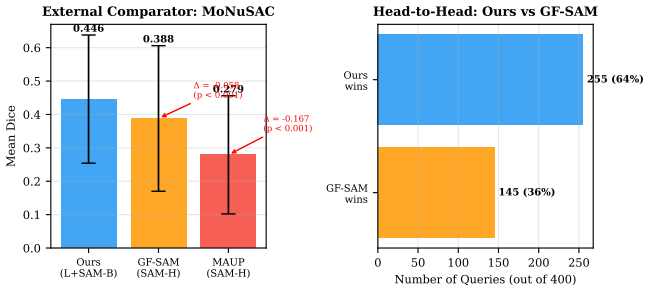


Figure 5: External comparator results. (a) Mean Dice with error bars. GF-SAM and MAUP underperform despite using SAM-ViT-H as the segmenter. (b) Head-to-head comparison: RAMP wins 255/400 queries against GF-SAM.

Table 8: Cross-dataset deployment envelope (trained on MoNuSAC SAM-B).

Dataset	Inst_AUC	Cat_AP	CFR_Red	ECE
MoNuSAC (holdout)	0.928	0.528	79.2%	0.027
MoNuSeg (transfer)	0.874	0.000	0.0%	0.060
IDRID (transfer)	0.994	0.997	6.0%	0.477

MoNuSeg: 0% catastrophic prevalence—no failure to detect.
IDRID: 80.7% catastrophic prevalence, ECE=0.477 (severe miscalibration).

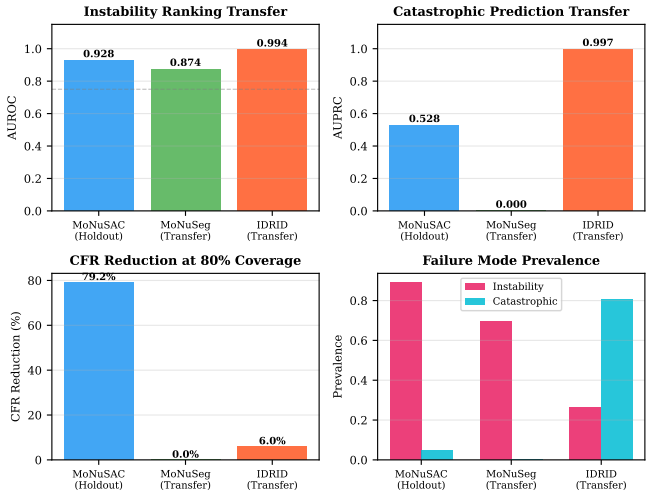


Figure 6: Characterizing the deployment envelope. (a) Instability ranking transfers well across datasets (AUROC 0.87–0.99). (b) Catastrophic prediction is prevalence-dependent. (c) CFR reduction varies with catastrophic prevalence. (d) The three datasets span the effective range [5%, 60%] of catastrophic prevalence.

catastrophic prevalence and strong predictive performance (CFR_Red = 79.2%).

These three points define a **deployment envelope**: the rank-based selective predictor is effective when the catastrophic failure prevalence falls in an intermediate range—roughly between 5% and 60%. Below this range, selective prediction is unnecessary because failures are too rare to justify abstention. Above this range, the prevalence is too high for the ranking signal to maintain calibration, and the predictor cannot reliably distinguish between the many failing episodes. This criterion can be estimated from a small unlabeled sample of the target dataset by running the reference-resampling protocol and measuring the fraction of episodes with mean Dice < 0.1, providing a practical screening tool for deployment decisions.

6 Discussion

6.1 The Structural Basis of the Anti-Correlation

The geometric account in Section 3 explains the anti-correlation through the entropy of the matching map, and three independent lines of evidence from our experiments converge on the same conclusion. The factorial ANOVA shows that the matcher has no effect on the anti-correlation ($p = 0.73$), placing the bottleneck downstream of feature matching, exactly where the entropy-driven point selection mechanism operates. The segmenter affects instability prevalence but not the anti-correlation structure—the pattern holds across all six segmenters tested, as predicted by the geometric model. Feature ablation reveals dissociation between f_4 (mask-level prediction Jaccard, dominating catastrophic detection) and f_1 (feature-level matching map disagreement, dominating instability detection). The two failure modes are encoded at different levels of the pipeline, reflecting the separation between entropy-driven point selection and downstream segmentation outcome. Taken together, these results establish that the anti-correlation is a structural property of the two-stage pipeline, not an artifact of a particular model or dataset.

6.2 Reliability vs. Sophistication in Prompt Construction

The finding that a simple top-K pipeline outperforms GF-SAM and MAUP—both of which use the larger SAM-ViT-H segmenter—requires careful interpretation. GF-SAM's graph-based point selection and MAUP's K-means uncertainty sampling are sophisticated strategies that excel on natural images with clear semantic structure. On dense nuclear images, however, their clustering mechanisms introduce brittleness: the semantic grouping they rely on is weak in histopathology, leading to unstable point selection. The frozen top-K approach makes no assumptions about semantic grouping—it simply selects the highest-confidence points. This is not an argument against sophisticated prompting in general. On their original benchmarks, GF-SAM and MAUP outperform simple baselines. The result is specific to dense, fine-grained segmentation tasks where precise point placement matters more than semantic clustering. The broader implication is that *prompt-construction reliability* is a dimension of evaluation orthogonal to *prompt quality*, and both should be reported when evaluating prompting methods.

The same pattern extends to the three domain-adapted methods. SAM-Med2D, despite medical fine-tuning, achieves Dice = 0.438 (0.008 below the simple pipeline). Its medical adapter layers improve feature extraction for medical images but do not address reference-induced failure—the fine-tuning adapts the model to the domain, not to the reference selection problem. CycleSAM's cycle-consistency constraint and H-SAM's hierarchical design transfer poorly to nuclear segmentation because they are optimized for spatial structures (surgical scenes) and textural patterns (document images) that are absent in histopathology.

6.3 Pipeline-Level vs. Model-Level Uncertainty

Pipeline-internal features substantially outperform model uncertainty methods for selective prediction in this setting. MC Dropout and Deep Ensemble estimate epistemic uncertainty in the segmenter's parameters, which is largely independent of reference-induced failure. The anti-correlation is a pipeline-level property that single-model uncertainty estimators cannot detect because they operate within a single stage of the pipeline. This result has practical implications: for visual prompting pipelines, investing in reference auditing—extracting features from multi-reference runs and training dual-model predictors—yields higher reliability gains than investing in model uncertainty estimation. The two approaches are complementary: model uncertainty could be incorporated as an additional feature in the dual-model predictor, potentially narrowing the oracle gap.

6.4 Practical Implications

The reference auditing approach has several consequences for the deployment of visual prompting pipelines. Because the rank-based design avoids per-model calibration thresholds, the selective predictor can be transferred across datasets without retraining. The deployment envelope criterion—estimating catastrophic prevalence from a small unlabeled sample—provides a practical screening tool: if the estimated prevalence falls below 5%, selective prediction is unnecessary; if it exceeds 60%, the method should not be deployed without additional adaptation. The finding that simple prompt construction is competitive on dense segmentation tasks suggests that practitioners should evaluate both prompt quality and prompt-construction reliability when selecting a prompting strategy. Finally, the dissociation between matcher-level and segmenter-level features demonstrates that pipeline-level analysis captures failure modes invisible to single-stage diagnostics.

6.5 Limitations

Several limitations bound the scope and generality of our findings. The full factorial audit covers MoNuSAC only, and the cross-dataset validation uses 30 queries per dataset, limiting statistical power for prevalence estimation. Nuclear and lesion segmentation are the only tasks evaluated; organ or tumor segmentation may exhibit different failure mode structures, particularly if the pipeline is applied to 3D volumes or multi-organ scenarios where the entropy characteristics of matching maps differ. The five external methods—GF-SAM, MAUP, CycleSAM, H-SAM, and SAM-Med2D—were designed for different domains, and their performance on our protocol may not reflect their capabilities on their original benchmarks. The oracle upper bound (77–89%) exceeds RAMP's performance, indicating that the current feature set and predictor architecture leave room for improvement. IDRID's ECE = 0.477 demonstrates that the rank-based design mitigates but does not eliminate calibration drift in extreme prevalence regimes; alternative ranking strategies, such as learning-to-rank approaches, may improve robustness. The four test-time-safe features require multiple reference runs; features requiring fewer references would improve practical usability. The deployment envelope criterion (prevalence in [5%, 60%]) is derived from only three datasets and should be validated on a broader range of tasks and modalities, including natural images, to establish its generality.

7 Conclusion

Reference-induced instability and catastrophic failure are structurally anti-correlated in two-stage visual prompting pipelines. This anti-correlation follows from the geometry of the point-prompt conversion bottleneck: the entropy of the matching map determines whether a query enters the low-entropy regime (high instability, low catastrophic risk) or the high-entropy regime (low instability, high catastrophic risk). The structure is universal across six segmenters and four matchers and is driven by the segmenter rather than the matcher. RAMP exploits this structure through a dual-model selective predictor, achieving 56–84% CFR reduction at 80% coverage on MoNuSAC while remaining calibration-free and transferable across datasets. A simple frozen top-K pipeline outperforms GF-SAM, MAUP, CycleSAM, H-SAM, and SAM-Med2D under identical conditions, demonstrating that prompt-construction reliability can outweigh prompt sophistication on dense segmentation tasks. By characterizing the deployment envelope through catastrophic failure prevalence, we identify the intermediate regime (roughly 5–60%) where rank-based selective prediction is effective, providing a practical screening criterion for deployment decisions that can be estimated from a small unlabeled target-domain sample.

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